

CombatHub & Combat Shop — Project Proposal

1. Executive Summary

CombatHub is a full-stack combat sports ecosystem that unifies social networking, e-commerce, content sharing, event management, and gym discovery into a single platform. The project consists of two interconnected single-page applications — CombatHub (social platform) and Combat Shop (marketplace) — served by a common Django REST Framework backend. It is a capstone project by Levi Mwangi.

2. The Problem

The combat sports community (fighters, coaches, gym owners, equipment vendors, and fans) is poorly served by existing digital platforms. The ecosystem is fragmented across multiple disconnected services:

- **Gear and equipment** — purchased on generic e-commerce sites with no combat-specific filtering (weight class, stance, sport discipline)
- **Coaching and gym discovery** — handled through separate directories or word-of-mouth
- **Community interaction** — dispersed across generic social media platforms with no role-based identity or combat-sport context
- **Event promotion** — scattered across social media event pages, flyers, and niche forums
- **Vendor management** — combat sports equipment vendors have no purpose-built platform to manage products, orders, and brand presence alongside their community engagement
- **Content moderation** — combat sports content (sparring footage, technique discussions) is often mis-flagged on generic platforms that don't understand the context

There is **no single platform** that:

1. Recognizes the distinct **roles** in combat sports (athlete, coach, gym owner, vendor) and provides appropriate tools for each
2. Combines a **social network** with a **purpose-built marketplace** for combat sports products

3. Offers **premium monetization** tailored to combat sports vendors (tiered plans based on product limits and features)
 4. Provides **SSO integration** allowing social platform users to seamlessly shop at the marketplace without creating separate accounts
 5. Includes **real-time messaging**, **gym profiles**, **event management**, and **content moderation** designed for the combat sports context
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3. The Problem Solved

CombatHub solves this fragmentation by delivering an all-in-one ecosystem that serves every stakeholder in the combat sports community:

For Athletes / Fighters

- **Social networking** — Post content (text, images, video), follow other athletes/coaches/gyms, like and comment on posts, bookmark content
- **Gear shopping** — Browse a marketplace with combat-specific filters (sport: boxing/BJJ/Muay Thai/MMA/fitness, weight class, stance), add to cart, checkout with card or M-Pesa
- **Gym discovery** — Find and browse gym profiles with location, facilities, and contact information
- **Event participation** — Discover and register for competitions, workshops, seminars, and tournaments
- **Messaging** — Real-time direct messaging with other platform users via WebSockets, including view-once images
- **Content discovery** — Search across users, posts, gallery, and products from a single search endpoint

For Coaches

- **Professional profile** — Showcase specialties, certifications, experience, and hourly rates
- **Content creation** — Create posts and share techniques, training footage, and educational content
- **Athlete discovery** — Find and follow athletes, view their profiles and activity
- **Event management** — Create and manage events (workshops, seminars) with participant registration
- **Group creation** — Create public or private groups for team/club management (premium feature)

For Gym Owners

- **Gym profile** — List gym with name, description, location, contact details, logo, and cover image
- **Facility promotion** — Feature gym listings for visibility
- **Event hosting** — Create and manage gym-hosted events
- **Member engagement** — Connect with members through the social platform

For Equipment Vendors

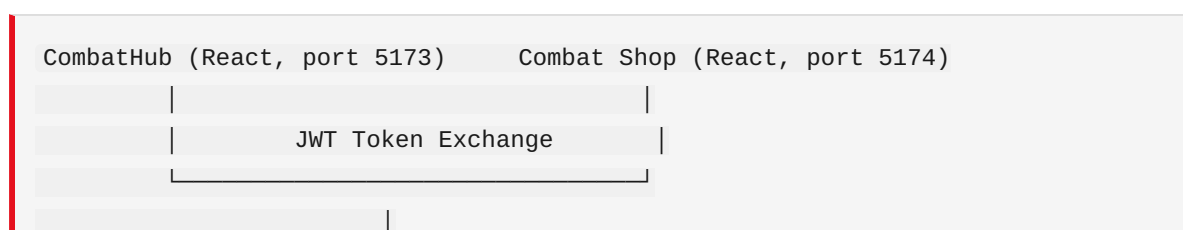
- **Product management** — Full CRUD for products with images, variants (size/color/material/weight), pricing, discounts, and stock tracking
- **Brand registration** — Register brands with admin approval workflow for credibility
- **Order management** — View and fulfill customer orders with confirmation workflow
- **Dashboard analytics** — Track product counts, order volumes, revenue, and follower counts
- **Premium subscription plans** — Scale from Free (5 products) to Basic (\$19/mo, 50 products) to Pro (\$49/mo, unlimited) with feature gating
- **Limited edition drops** — Create time-limited product drops with countdowns and notification systems

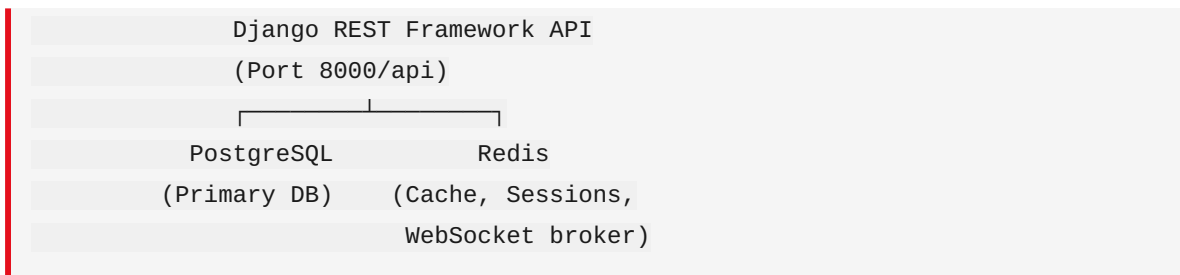
For Platform Administrators

- **User management** — View all users, change roles, block messaging, manage user accounts
- **Brand approval** — Review and approve/revoke brand registrations
- **Content moderation** — Auto-flagging of profanity, hate speech, violence, and spam with review workflow
- **Reporting system** — User-submitted content reports with status tracking (pending/reviewed/dismissed/actioned)

4. Architecture

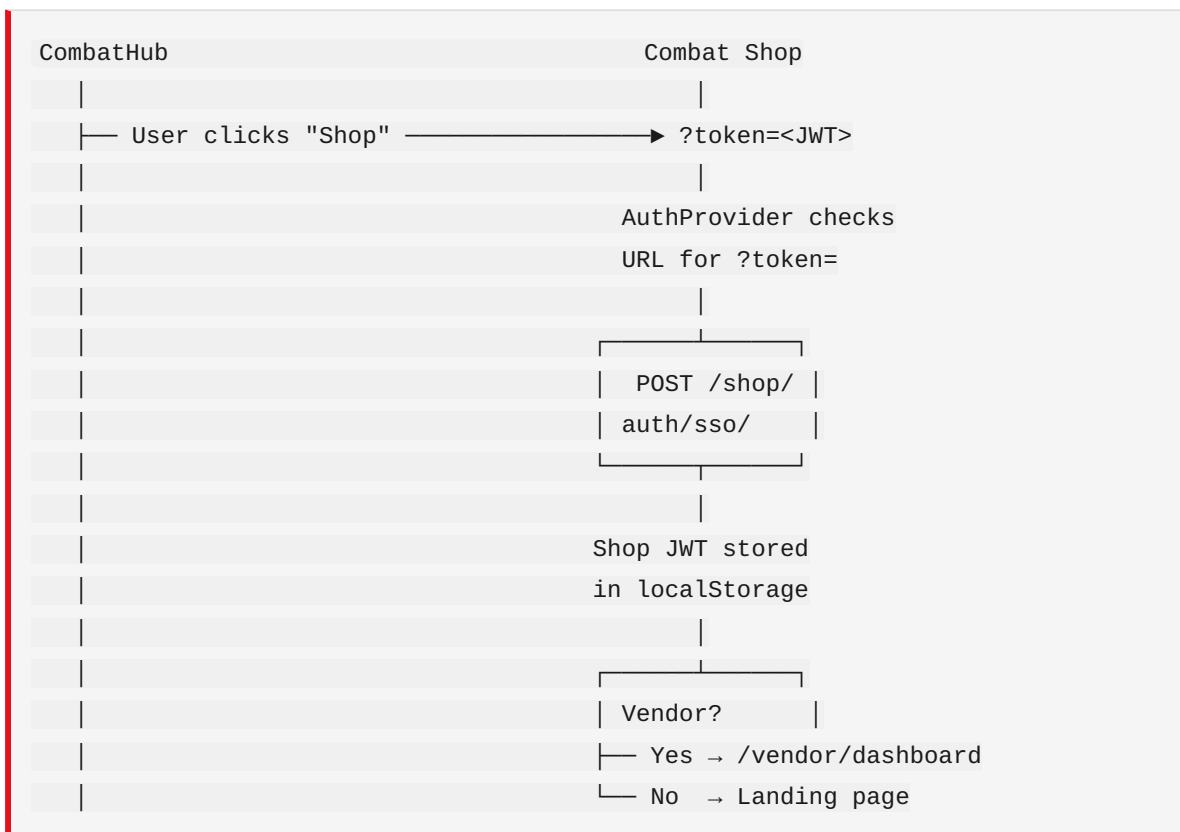
Platform Overview





The platform uses a **dual-frontend, single-backend** architecture: - **CombatHub** (port 5173) — Social networking SPA for the combat sports community - **Combat Shop** (port 5174) — Standalone marketplace SPA with customer storefront and vendor dashboard - **Django REST Framework** (port 8000) — Shared backend serving both frontends via REST APIs and WebSocket connections

SSO Integration



- SSO processes **before** any existing token check — no stale vendor session race
 - Sync logout via `shop_active` cookie (`domain=localhost`) — checked on tab focus via `visibilitychange` event
 - Cookie is set after login/SSO with 24h max-age, cleared on CombatHub logout
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5. Technology Stack

Layer	Technology
Backend	Django 6.x, Django REST Framework 3.17+, Python 3.12+
Frontend (CombatHub)	React 19, Vite 8, Tailwind CSS 4, Framer Motion 12, GSAP 3
Frontend (Combat Shop)	React 19, Vite 8, Tailwind CSS 4, Recharts
Database	PostgreSQL 16+
Cache & Session Store	Redis 7+ (falls back to local memory)
Real-time	Django Channels 4.3+ (WebSockets), Daphne 4.2+ (ASGI server)
Auth	SimpleJWT (access + refresh tokens with rotation), Google OAuth
Task Queue	Celery 5.6+ (with Redis broker)
Payments	Stripe 15.1, Africa's Talking 2.0 (SMS)
Email	SMTP (Gmail), SendGrid
Media	Cloudinary, Pillow
Maps	Leaflet + react-leaflet
Containerization	Docker + Docker Compose (multi-stage builds)
Deployment	Railway, Render, DigitalOcean, AWS, Fly.io

6. App Structure & Code Organization

6.1 Backend Django Apps (13 apps)

```
backend/
├─ accounts/           # Core: users, auth, social features
├─ products/           # E-commerce: products, cart, orders
├─ shop/               # Vendor management, premium, brands
├─ events/             # Event management
├─ gyms/               # Gym profiles
├─ reviews/           # Product and gym reviews
├─ subscriptions/     # Newsletter subscriptions
├─ coaches/           # Coach profiles
├─ exercises/         # Exercise library
└─ chat/              # WebSocket consumers
```

└─ comments/	# Generic commenting system
└─ common/	# Shared permissions, pagination
└─ payments/	# Payment processing (stub)
└─ config/	# Django project configuration

6.2 Data Models (40+ models)

accounts app — Core user system: - `User` — Custom user model with email auth, 5 roles (athlete/coach/gym_owner/vendor/admin), Google OAuth, TOTP 2FA, email verification - `Profile` — One-to-one with User: bio, avatar, phone, weight_class, height, reach, stance, business info, premium tracking, last_seen - `Post` / `PostLike` / `PostComment` / `PostCommentLike` — Social posts with like/dislike voting and threaded comments - `GalleryItem` / `GalleryLike` / `GalleryComment` — Image gallery with likes and threaded comments - `Follow` — Follower/following relationships - `Message` — Direct messages with view-once image support - `Group` / `GroupMember` / `GroupMessage` — Public/private groups with invite system - `Notification` — 12 notification types (follow, profile update, new post, premium events, etc.) - `Bookmark` — Save posts, gallery items, and products - `Report` — Content reporting with status workflow - `ContentFlag` — Auto-moderation flagging with regex-based detection - `BlockedUser` — User blocking - `IPLog` — SHA-256 hashed IP tracking on every request - `EmailVerificationCode` / `PhoneVerificationCode` — Verification code storage - `VendorAccessCode` — Access codes for coach/gym_owner registration - `PaymentInfo` — Stored payment methods (M-Pesa, card) - `SiteContent` — Dynamic site content pages - `UsernameChange` — Username change history (max 2/month)

products app — Marketplace: - `Product` — Full product model: name, description, brand, category, sport_tags (JSON), images (JSON), price, stock, limited edition, serial number, drop date, featured, vendor, is_visible, discount_active, discount_percent; has `effective_price` computed property - `Category` — Hierarchical categories with sport_tag and ordering - `ProductVariant` — Size, color, material, weight_oz, stock, price_override - `Drop` — Limited edition drops with launch_time, max_quantity, countdown - `DropNotification` — User drop notifications - `Cart` / `CartItem` — Shopping cart with quantity management - `Order` — Full order tracking: items (JSON), subtotal, total, status (pending/confirmed/paid/shipped/delivered/cancelled), payment method, shipping address, payment intent - `Favorite` — User product favorites - `ProductComment` — Product reviews with threaded replies

shop app — Vendor management: - `VendorPremiumPlan` — Tiered plans: Free (\$0/5 products/no discounts), Basic (\$19/mo/50 products/discounts), Pro (\$49/mo/unlimited/discounts+analytics) - `VendorProfile` — Business name, location, description, lat/lng, avatar - `VendorSubscription` — Tracks plan subscription with status (active/trialing/canceled/expired), 14-day trial period - `VendorInvitation` — Invitation system with UUID tokens - `Brand` — Brand catalogue with admin approval workflow

events app: - `Event` — Events with type (competition/workshop/seminar/tournament/other), location, dates, max participants, entry fee, poster - `EventParticipant` — Registration tracking

gyms app: - `Gym` — Gym profiles with owner, contact details, location, logo, cover image, featured flag

reviews app: - `ProductReview` — Product ratings and reviews

subscriptions app: - `NewsletterSubscriber` — Email newsletter subscriptions

6.3 API Endpoints (100+ endpoints)

Authentication (`/api/auth/`): - `register/` — User registration with email validation (DNS MX check) - `login/` — Email or username login; blocks username login for vendor/coach/gym_owner - `verify-login/` — TOTP 2FA verification step - `google/` — Google OAuth sign-in/sign-up - `verify-access-code/` — Access code verification for coach/gym_owner - `password-reset/` / `password-reset/confirm/` — Password reset flow - `refresh/` — JWT token refresh (rotation + blacklist) - `2fa/setup/` / `2fa/verify/` / `2fa/disable/` — TOTP management

Users (`/api/auth/`): - `me/` — Current user profile (GET, PATCH) - `users/` — User listing with search, filter by role - `users/<id>/` — Public profile view - `users/<id>/follow/` / `unfollow/` — Follow/unfollow - `users/<id>/block/` — Block user - `users/<id>/remove-follower/` — Remove follower

Social (`/api/auth/`): - `posts/` — Post CRUD with like/dislike, comments - `gallery/` — Gallery CRUD with likes, comments - `search/` — Cross-resource search (users, posts, gallery, products) - `bookmarks/` / `bookmarks/toggle/` — Bookmark management - `notifications/` — Notification listing, unread count, mark read - `reports/` — Content reporting - `conversations/` — Message conversations - `conversations/<id>/send/` — Send message - `messages/<id>/edit/` / `delete/` / `view-once/` — Message management

Groups (`/api/auth/`): - `groups/` — CRUD with join/leave, invite, member management - `groups/<id>/messages/` / `messages/send/` — Group messaging

Premium (`/api/auth/`): - `premium/start-trial/` — 30-day CombatHub premium trial - `premium/check/` — Premium status check - `premium/setup-payment/` — Payment setup - `premium/cancel/` — Cancel premium

Admin (`/api/auth/`): - `admin/users/` — List all users - `admin/users/<id>/role/` — Change user role - `admin/users/<id>/toggle-messaging-block/` — Block/unblock messaging

Products (`/api/`): - `products/` — Product listing with filters (category, brand, sport, price, featured, limited) - `products/<id>/` — Product detail - `products/<id>/favorite/` —

Toggle favorite - `categories/` — Category listing - `drops/active/` — Active limited drops
- `drops/<id>/notify/` — Drop notification signup - `cart/` / `cart/add/` / `cart/items/`
`<id>/` — Cart management - `checkout/` — Order creation - `orders/` / `orders/<id>/` —
Order history

Shop (`/api/shop/`): - `shop/auth/login/` — Vendor-only login - `shop/auth/sso/` — SSO
token exchange - `shop/auth/activate/` — Vendor invitation activation - `shop/dashboard/`
— Vendor dashboard stats - `shop/products/` — Vendor product CRUD - `shop/products/`
`<id>/toggle-discount/` — Discount toggle - `shop/products/<id>/toggle-visibility/` —
Visibility toggle - `shop/upload/` — Product image upload - `shop/orders/` — Vendor order
listing - `shop/orders/<id>/confirm/` — Order confirmation - `shop/premium/plans/` —
Premium plan listing - `shop/premium/subscription/` — Current subscription - `shop/`
`premium/subscribe/` — Subscribe to plan - `shop/premium/cancel/` — Cancel subscription
- `shop/brands/` — Brand CRUD - `shop/admin/brands/` — Admin brand listing - `shop/`
`admin/brands/<id>/toggle-approval/` — Brand approval toggle

Events (`/api/events/`): - Event CRUD with register/unregister

Gyms (`/api/gyms/`): - Gym CRUD

Reviews (`/api/reviews/`): - Product review create/list

6.4 Permission System

```
common/permissions.py:
```

```
IsAdmin          - role == 'admin'
IsCoach           - role == 'coach'
IsVendor          - role == 'vendor'
IsSeller          - role in (vendor, coach, gym_owner)
IsVendorOrReadOnly - read: anyone; write: vendor
IsPremium         - authenticated + is_premium_active()
IsPremiumOrHigher - same as IsPremium
```

```
shop/views.py:
```

```
HasVendorPremium - authenticated + vendor + active VendorSubscription
```

6.5 WebSocket Consumers

- **ChatConsumer** (`ws/chat/<user_id>/`) — Real-time direct messaging with typing indicators, read/delivered receipts, view-once images
- **NotificationConsumer** (`ws/notifications/`) — Real-time notification stream (new messages, follows)
- Rooms: `chat_{sorted_ids}` for DMs, `user_{id}_notifications` for notifications

6.6 Celery Tasks

- `send_email_async` — Generic async email sending
- `send_welcome_email_async` — Welcome email on registration
- `send_access_code_email_async` — Access code email
- `create_notifications_async` — Bulk notification creation
- `log_event_async` — Async logging

6.7 Signal System

- **accounts/signals.py**: Profile creation on user signup, welcome email, content auto-flagging (profanity/hate speech/violence/spam), follower notifications for new posts/gallery, cache invalidation, custom signals (`content_flagged` , `premium_activated` , `premium_expired` , `order_completed` , `user_registered`)
- **products/signals.py**: Cache invalidation on product save/delete
- **gyms/signals.py**: Cache invalidation on gym save/delete
- **events/signals.py**: Cache invalidation on event/participant save/delete
- **reviews/signals.py**: Cache invalidation on review save/delete
- **subscriptions/signals.py**: Cache invalidation on subscribe/unsubscribe

6.8 Frontend Structure

CombatHub (`frontend/src/`): - **Pages** (45+): Home, Login, Register, VerifyLogin, VerifyAccessCode, TotpSetup, UsernameSetup, EmailVerify, ForgotPassword, ResetPassword, Settings, ProfileEdit, PublicProfile, Forum, PostDetail, CreatePost, Gallery, GalleryDetail, Community, Messages, Groups, GroupDetail, Events, EventDetail, EventForm, MyEvents, Shop, ProductDetail, Cart, Checkout, OrderHistory, OrderDetail, VendorDashboard, VendorProductForm, VendorAbout, SellerOrders, PaymentSetup, Premium, GymOwnerDashboard, CoachDashboard, AdminPage, About, ContactUs, Help, Terms, CommunityGuidelines, Newsletter - **Layouts**: MainLayout (header + sidebar), AuthLayout - **Providers**: AuthProvider (JWT + user state), CartProvider, ThemeProvider - **Components**: Button, Input, Spinner, Skeleton, Toast, Modal, EventCard, GoogleLoginWrapper, Reveal, ReportModal, EmojiPicker, PremiumBadge, MapPicker

Combat Shop (`shop/src/`): - **Pages** (30): Landing, Login, Activate, AccessGate, Shop, Products, ProductList, ProductView, ProductForm, Cart, Checkout, Orders, OrderDetail, OrderHistory, CustomerOrderDetail, Dashboard, Categories, Brands, Settings, CustomerSettings, VendorPremium, VendorEvents, Gallery, Contact - **Layouts**: ShopLayout (vendor sidebar + header), CustomerLayout (customer nav) - **Providers**: AuthProvider (SSO + vendor login), CartProvider, ThemeProvider - **Components**: Icons.jsx (30+ SVG icon components)

7. Key Features in Detail

7.1 Role-Based Access Control

Five user roles with distinct capabilities enforced at both API (permission classes) and UI (route guards, conditional rendering) levels: - **Athlete**: Browse marketplace, cart, checkout, social posting, messaging - **Coach**: Create posts (premium), groups (premium), events, coach profile - **Gym Owner**: Gym management, events, groups, vendor access - **Vendor**: Product CRUD, order management, brand registration, premium plans - **Admin**: User management, brand approval, content moderation

7.2 Vendor Premium Subscription System

Three-tier plan model: | Plan | Price | Max Products | Discounts | Analytics |
|-----|-----|-----|-----|-----| | Free | \$0 | 5 | No | No | | Basic | \$19/mo | 50 |
Yes | No | | Pro | \$49/mo | Unlimited | Yes | Yes |

- Auto-assigned to Free plan on first check
- 14-day free trial on paid plans (status='trialing')
- `HasVendorPremium` gates: product creation, editing, deletion, image upload, discount toggling
- Subscription lifecycle: trialing → active → canceled/expired

7.3 Content Moderation

- Auto-flagging via regex patterns for profanity, hate speech, violence, spam
- Flagged content status: pending → auto_moderated/approved/dismissed
- User reporting system with reason categories
- IP logging with SHA-256 hashing

7.4 Combat-Specific Taxonomies

- Sport tags: boxing, BJJ (Brazilian Jiu-Jitsu), Muay Thai, MMA (Mixed Martial Arts), fitness
- Weight classes, stances (orthodox/southpaw/switch), reach, height
- Products filterable by all combat-specific dimensions

7.5 Security

- JWT with access token (1h) + refresh token (7d) rotation and blacklisting
- TOTP two-factor authentication

- Email verification (6-digit codes)
- Phone verification via Africa's Talking
- Rate limiting: 100k/day authenticated, 120/hr anonymous, 3/hr register, 5/hr subscribe
- CORS whitelisting
- SHA-256 hashed IP tracking
- Access codes for coach/gym_owner registration
- Content flagging and reporting

7.6 Design System (Combat Shop)

- Liquid-glass cards with `backdrop-filter: blur()`
 - Nike red (#e5101d) accent color on dark theme
 - CSS variable-based light/dark mode toggling
 - Desktop sidebar + mobile drawer for vendor navigation
 - Animated transitions (slideUp, slowZoom, fadeIn, scaleIn)
 - Background music auto-play from MP3 assets
 - Hero video carousel with Pexels fallback
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8. Target Audience

User Role	Needs Served
Athletes / Fighters	Social networking, gear shopping, gym discovery, event participation, messaging
Coaches	Profile showcasing, athlete discovery, service promotion, event management
Gym Owners	Gym listing, facility promotion, member recruitment, event hosting
Equipment Vendors	Product sales, brand management, order fulfillment, premium scaling
Combat Sports Fans	Content discovery, community engagement, event attendance
Platform Administrators	User management, brand approval, content moderation

9. Scope & Feasibility

The project is fully implemented and functional:

- **Backend:** 13 Django apps, 40+ database models, 100+ API endpoints, WebSocket consumers, Celery tasks, signal handlers, comprehensive permission system
- **CombatHub Frontend:** 45+ pages, full routing, auth flows, messaging, groups, marketplace, premium, admin tools
- **Combat Shop Frontend:** 24 pages, customer storefront, vendor dashboard, SSO integration, cart/checkout, premium plans, brand management, gallery
- **Testing:** Both frontends build successfully with zero errors

Running Locally

```
# Terminal 1: Backend
cd backend && python manage.py runserver

# Terminal 2: CombatHub
cd frontend && npm run dev

# Terminal 3: Combat Shop
cd shop && npm run dev
```

Or with Docker: `docker compose up --build -d`

10. Deployment

The project includes Docker configuration for production deployment: - **Backend:** Python 3.12 + Daphne (ASGI) with automated migrations, static collection, and server startup - **Frontend:** Multi-stage Docker build (Node build → Nginx serve) with API reverse proxy - **Supporting services:** PostgreSQL and Redis - **Platform targets:** Railway, Render, DigitalOcean App Platform, AWS ECS, Fly.io

11. Conclusion

CombatHub addresses a genuine gap in the digital landscape for combat sports. By uniting social networking, e-commerce, event management, and professional profiles into a single ecosystem with role-based access, it provides the combat sports community with a purpose-built platform that no existing solution offers. The two-application architecture with SSO integration demonstrates sophisticated web development patterns while delivering

practical value to a passionate and underserved user base. With a comprehensive backend API, two fully-featured frontend applications, real-time communication, content moderation, and a tiered premium monetization system, CombatHub is a complete, production-ready platform.